DEFENCE JOURNEY

NDA Mathematics Notes

Complete formula and concept notes for the UPSC NDA Maths paper

NDA | MATHS

NDA Mathematics carries **300 marks** (120 questions, 2.5 marks each, 1/3 negative marking). These notes cover every unit of the UPSC syllabus with the formulas and shortcuts you need most. Learn the formulas first, then practise from the Mock Tests.

Unit	Approx. weight	Priority
Algebra (sets, complex numbers, quadratic, sequences, P&C, binomial, matrices)	~20 questions	High
Trigonometry (identities, heights & distances, inverse)	~20 questions	High
Analytical Geometry 2D & 3D	~20 questions	High
Differential Calculus	~15 questions	High
Integral Calculus & Differential Equations	~20 questions	High
Vectors	~8 questions	Medium
Statistics & Probability	~15 questions	Medium

TIP: Question counts are approximate and change every year. Calculus and Trigonometry give the most reliable marks.

1. Sets, Relations and Functions

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- A set with n elements has $\mathbf{2^n}$ subsets and $2^n - 1$ non-empty subsets.
- De Morgan: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.
- A relation is an **equivalence relation** if it is reflexive, symmetric and transitive.
- A function is **one-one** if different inputs give different outputs, **onto** if every element of the codomain is used.
- If A has m elements and B has n elements, the number of functions $A \to B$ is n^m.

2. Complex Numbers

- $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Divide the power by 4 and use the remainder.
- For $z = a + ib$: modulus $|z| = \sqrt{(a^2 + b^2)}$, conjugate $\bar{z} = a - ib$, and $z \cdot \bar{z} = |z|^2$.
- Cube roots of unity: 1, ω, ω^2 where $\omega^3 = 1$ and $\mathbf{1 + \omega + \omega^2 = 0}$.
- Polar form: $z = r(\cos\theta + i \sin\theta)$. Product: multiply moduli, add arguments.

3. Quadratic Equations

For $ax^2 + bx + c = 0$:

- Roots: $x = [-b \pm \sqrt{(b^2 - 4ac)}] / 2a$. Discriminant $D = b^2 - 4ac$.
- Sum of roots $= -b/a$. Product of roots $= c/a$.
- $D > 0$: two real distinct roots. $D = 0$: equal real roots. $D < 0$: complex roots (conjugate pair).
- Equation with roots α and β: $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.
- $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.

4. Sequences and Series

	Arithmetic Progression (AP)	Geometric Progression (GP)
n-th term	$a + (n - 1)d$	$a \cdot r^{n-1}$
Sum of n terms	$(n/2)[2a + (n - 1)d] = (n/2)(a + l)$	$a(r^n - 1)/(r - 1)$ for $r \neq 1$
Sum to infinity	—	$a/(1 - r)$ when $\lvert r\rvert < 1$
Mean of a, b	$(a + b)/2$	$\sqrt{(ab)}$

- Sum of first n natural numbers $= n(n+1)/2$. Sum of squares $= n(n+1)(2n+1)/6$. Sum of cubes $= [n(n+1)/2]^2$.
- For positive numbers, AM ≥ GM ≥ HM.

5. Logarithms

- $\log(mn) = \log m + \log n$. $\log(m/n) = \log m - \log n$. $\log m^p = p \log m$.
- Change of base: $\log_a b = \log b / \log a$. Also $\log_a b \times \log_b a = 1$.
- $\log_a 1 = 0$, $\log_a a = 1$. Logarithm is defined only for positive numbers.

6. Permutations, Combinations and Binomial Theorem

- ${}^nP_r = n!/(n - r)!$ and ${}^nC_r = n!/[r!(n - r)!]$
- ${}^nC_r = {}^nC_{n-r}$. ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.
- Arrangements of n things where p are alike of one kind and q alike of another: $n!/(p!\, q!)$.
- Circular arrangements of n distinct things $= (n - 1)!$.
- $(a + b)^n$ has $n + 1$ terms. General term $T_{r+1} = {}^nC_r\, a^{n-r}\, b^r$.
- Sum of all binomial coefficients $= 2^n$.

7. Matrices and Determinants

- Matrix product AB exists only if columns of A = rows of B. In general $AB \neq BA$.
- $(AB)^T = B^T A^T$. $(AB)^{-1} = B^{-1}A^{-1}$.
- $A^{-1} = \text{adj}(A)/|A|$, and it exists only when $|A| \neq 0$ (non-singular matrix).
- $|AB| = |A||B|$. For an $n \times n$ matrix, $|kA| = k^n|A|$.
- 2×2 determinant: $|a\ b;\ c\ d| = ad - bc$.
- Swapping two rows changes the sign of the determinant. Two identical rows make it zero.
- System $AX = B$ has a unique solution if $|A| \neq 0$. For a homogeneous system, $|A| = 0$ means non-trivial solutions exist.

8. Trigonometry

Standard values

θ	0°	30°	45°	60°	90°
sin θ	0	1/2	1/√2	√3/2	1
cos θ	1	√3/2	1/√2	1/2	0
tan θ	0	1/√3	1	√3	not defined

Formulae

- $\sin^2\theta + \cos^2\theta = 1$. $1 + \tan^2\theta = \sec^2\theta$. $1 + \cot^2\theta = \operatorname{cosec}^2\theta$.
- $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$. $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$.
- $\tan(A \pm B) = (\tan A \pm \tan B)/(1 \mp \tan A \tan B)$.
- $\sin 2A = 2 \sin A \cos A$. $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$.
- $\tan 2A = 2 \tan A/(1 - \tan^2 A)$.
- $\sin 3A = 3 \sin A - 4 \sin^3 A$. $\cos 3A = 4\cos^3 A - 3 \cos A$.
- $\sin C + \sin D = 2 \sin[(C+D)/2] \cos[(C-D)/2]$. $\cos C + \cos D = 2 \cos[(C+D)/2] \cos[(C-D)/2]$.
- Quadrant rule (ASTC): All positive in Q1, Sine in Q2, Tangent in Q3, Cosine in Q4.
- $\sin(90° - \theta) = \cos\theta$, $\sin(180° - \theta) = \sin\theta$, $\cos(180° - \theta) = -\cos\theta$, $\tan(180° - \theta) = -\tan\theta$.

Triangles

- Sine rule: $a/\sin A = b/\sin B = c/\sin C = 2R$.
- Cosine rule: $\cos A = (b^2 + c^2 - a^2)/2bc$.
- Area = ½ab sin C. Area (Heron) = $\sqrt{[s(s-a)(s-b)(s-c)]}$ where $s = (a+b+c)/2$.

Heights and Distances

Angle of elevation is measured upward from the horizontal, angle of depression downward. Draw a right-angled triangle and use tan θ = height/base. If the angle of elevation of the top of a tower is 45°, then height = distance from the base.

Inverse trigonometry

- $\sin^{-1}x + \cos^{-1}x = \pi/2$. $\tan^{-1}x + \cot^{-1}x = \pi/2$.
- Principal values: $\sin^{-1}x \in [-\pi/2, \pi/2]$, $\cos^{-1}x \in [0, \pi]$, $\tan^{-1}x \in (-\pi/2, \pi/2)$.
- $\tan^{-1}x + \tan^{-1}y = \tan^{-1}[(x + y)/(1 - xy)]$ when $xy < 1$.

9. Analytical Geometry of Two Dimensions

- Distance between (x_1, y_1) and $(x_2, y_2) = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$.
- Section formula (internal, ratio m:n): $[(mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n)]$. Mid-point uses m = n = 1.
- Slope $m = (y_2 - y_1)/(x_2 - x_1) = \tan\theta$. Parallel lines: $m_1 = m_2$. Perpendicular lines: $m_1 m_2 = -1$.
- Line equations: $y = mx + c$; $y - y_1 = m(x - x_1)$; $x/a + y/b = 1$.

- Angle between two lines: $\tan\theta = |(m_1 - m_2)/(1 + m_1 m_2)|$.
- Distance of point (x_1, y_1) from line $ax + by + c = 0$ is $|ax_1 + by_1 + c|/\sqrt{(a^2 + b^2)}$.
- Area of triangle with vertices (x_1,y_1), (x_2,y_2), $(x_3,y_3) = \frac{1}{2}|x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)|$.
- Centroid $= [(x_1+x_2+x_3)/3, (y_1+y_2+y_3)/3]$.

Circle

- Centre (h, k), radius r: $(x - h)^2 + (y - k)^2 = r^2$.
- General form $x^2 + y^2 + 2gx + 2fy + c = 0$ has centre $(-g, -f)$ and radius $\sqrt{(g^2 + f^2 - c)}$.
- A tangent is perpendicular to the radius at the point of contact.

Conics

Conic	Standard form	Key facts
Parabola	$y^2 = 4ax$	Focus (a, 0), directrix $x = -a$, latus rectum $= 4a$, $e = 1$
Ellipse	$x^2/a^2 + y^2/b^2 = 1$ $(a > b)$	$e = \sqrt{(1 - b^2/a^2)} < 1$, foci $(\pm ae, 0)$, major axis 2a, minor axis 2b
Hyperbola	$x^2/a^2 - y^2/b^2 = 1$	$e = \sqrt{(1 + b^2/a^2)} > 1$, foci $(\pm ae, 0)$, asymptotes $y = \pm(b/a)x$

10. Analytical Geometry of Three Dimensions

- Distance between two points $= \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]}$.
- Direction cosines l, m, n satisfy $\mathbf{l^2 + m^2 + n^2 = 1}$.
- Direction ratios a, b, c give direction cosines $a/\sqrt{(a^2+b^2+c^2)}$ etc.
- Two lines are perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$.
- Plane: $ax + by + cz + d = 0$, with (a, b, c) the direction ratios of its normal.
- Sphere with centre (a, b, c) and radius r: $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$.

11. Differential Calculus

Limits

- $\lim (\sin x)/x = 1$ and $\lim (\tan x)/x = 1$ as $x \to 0$.
- $\lim (1 + 1/x)^x = e$ as $x \to \infty$. $\lim (e^x - 1)/x = 1$. $\lim (a^x - 1)/x = \ln a$.
- For a 0/0 form, factorise, rationalise, or use L'Hôpital's rule (differentiate numerator and denominator separately).

Derivatives to remember

f(x)	f′(x)	f(x)	f′(x)
x^n	$n\,x^{n-1}$	e^x	e^x
sin x	cos x	ln x	1/x
cos x	−sin x	a^x	$a^x \ln a$
tan x	$\sec^2 x$	$\sin^{-1} x$	$1/\sqrt{(1-x^2)}$
cot x	$-\mathrm{cosec}^2 x$	$\cos^{-1} x$	$-1/\sqrt{(1-x^2)}$
sec x	sec x tan x	$\tan^{-1} x$	$1/(1+x^2)$
cosec x	−cosec x cot x	√x	1/(2√x)

- Product rule: (uv)′ = u′v + uv′. Quotient rule: (u/v)′ = (u′v − uv′)/v^2. Chain rule: d/dx f(g(x)) = f′(g(x))·g′(x).
- **Maxima and minima:** solve f′(x) = 0. If f″(x) < 0 it is a maximum, if f″(x) > 0 it is a minimum.
- f is increasing where f′(x) > 0 and decreasing where f′(x) < 0.
- Slope of the tangent at a point = f′(x) at that point. Slope of the normal = −1/f′(x).
- A function is continuous at x = a if left limit = right limit = f(a). Differentiable ⇒ continuous, but not the other way round (e.g. |x| at 0).

12. Integral Calculus and Differential Equations

Function	Integral	Function	Integral
x^n	$x^{n+1}/(n+1) + C$ $(n \neq -1)$	sin x	−cos x + C
1/x	ln|x| + C	cos x	sin x + C
e^x	$e^x + C$	$\sec^2 x$	tan x + C
a^x	$a^x/\ln a + C$	$1/(1+x^2)$	$\tan^{-1} x + C$
$1/\sqrt{(1-x^2)}$	$\sin^{-1} x + C$	$\mathrm{cosec}^2 x$	−cot x + C

- Integration by parts: $\int u\,v\,dx = u\int v\,dx - \int[u' \int v\,dx]\,dx$. Choose u in the order **ILATE** (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential).
- Substitution: if the integrand contains f(x) and f′(x), put t = f(x).
- Definite integral properties: $\int_a^b f(x)dx = -\int_b^a f(x)dx$. $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$.
- $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$ if f is even, and = 0 if f is odd.
- Area under y = f(x) between x = a and x = b is $\int_a^b |f(x)|\,dx$.
- **Differential equations:** order = highest derivative, degree = its power (after clearing roots and fractions). Solve variable separable equations by writing f(y)dy = g(x)dx and integrating both sides. A general solution has as many arbitrary constants as the order.

13. Vectors

- Dot product: a·b = |a||b| cos θ. i·i = j·j = k·k = 1, i·j = j·k = k·i = 0. If a·b = 0 the vectors are perpendicular.

- Cross product: $|a \times b| = |a||b| \sin\theta$, direction by the right-hand rule. $a \times b = -(b \times a)$. If $a \times b = 0$ the vectors are parallel.
- Area of a parallelogram with adjacent sides a, b = $|a \times b|$. Area of a triangle = $\frac{1}{2}|a \times b|$.
- Scalar triple product $[a\ b\ c] = a \cdot (b \times c)$ gives the volume of the parallelepiped. Vectors are coplanar if it is zero.
- Unit vector along a = $a/|a|$. Projection of a on b = $(a \cdot b)/|b|$.
- Position vector of the point dividing AB in ratio m:n internally = $(m \cdot b + n \cdot a)/(m + n)$.

14. Statistics and Probability

- Mean = $\Sigma x/n$. Median = middle value of sorted data. Mode = most frequent value.
- Empirical relation: **Mode = 3 Median − 2 Mean**.
- Variance $\sigma^2 = \Sigma(x - \text{mean})^2/n = \Sigma x^2/n - (\text{mean})^2$. Standard deviation $\sigma = \sqrt{\text{variance}}$.
- Adding a constant to every value changes the mean but not the standard deviation. Multiplying every value by k multiplies the standard deviation by $|k|$.
- Probability = favourable outcomes / total outcomes. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. $P(A') = 1 - P(A)$.
- Conditional probability: $P(A|B) = P(A \cap B)/P(B)$. Independent events: $P(A \cap B) = P(A) \cdot P(B)$.
- Bayes' theorem: $P(A_i|E) = P(A_i)P(E|A_i) / \Sigma P(A_j)P(E|A_j)$.
- Binomial distribution: $P(X = r) = {}^{n}C_{r}\, p^{r}\, q^{n-r}$, with mean np and variance npq.
- Throwing two dice gives 36 outcomes. P(sum = 7) = 6/36 = 1/6.

Exam Strategy for Maths

- Each question carries 2.5 marks and a wrong answer loses 0.83. Attempt only when you can eliminate at least two options.
- Many questions can be solved by putting in the option values rather than solving fully.
- Memorise squares up to 30, cubes up to 15, and the standard trigonometric values.
- Practise 30 to 40 mixed questions daily with a timer. Aim for under 1.2 minutes per question.
- Revise the formula sheet in these notes every morning for the last 30 days.